

Personal Exploration Onramp

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This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle*. Plots were typeset using *Matplotlib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

It's time to enjoy a break and consider what interests you in physical organic chemistry and what you would like to explore. Is there something in your honours project that you could dive into? Or perhaps a mystery that you encounter in your reading?

The Personal Exploration

As we progress through the course, it is time for you to start a personal relationship with a literature topic. You should choose a body of work and teach yourself and your peers about it. Along the way, you will write a personal exploration about your investigations. It is not a formal document but a narrative of your thinking, criticisms and appreciation of the work. It should read like a good story.¹

Consider the exploration documents from recent class meetings where I presented a topic related to the class meeting material. Observe that a paper is dissected and its conclusions analyzed by re-investigating the data presented. The data is plotted, and the results are compared to the author's. In many cases, errors were found. I usually seek results and data from other papers to help with the analysis. In a nutshell, I read a paper and then answered my own questions.

Read the literature and find a few papers that you are interested in. Pick a paper that has data to analyze. Your exploration must include some data analysis. You don't have to stay with the first paper you choose; you just have to start on your journey.

¹ Don't be too casual in your writing. The use of the word "dude" is frowned upon. Read the available examples, and you will see that the writing style is fairly formal, but I chose to use an active voice. This writing style is more personal and engaging than the typical passive voice favoured in publications.

Oh, The Places You Will Go

Where does one look to find papers that feature physical organic chemistry? Most journals that cover the field of organic chemistry will be a fertile hunting ground. Below are a few suggestions.

- [ACS Journal of the American Chemical Society](#)²
- [ACS Organic letters](#)³
- [ACS Journal of Organic Chemistry](#)⁴
- [Canadian Journal of Chemistry](#)⁵
- [Journal of Physical Organic Chemistry](#)⁶
- [Bio-organic Chemistry](#)⁷

² <https://pubs.acs.org/journal/jacsat>

³ <https://pubs.acs.org/journal/orlef7>

⁴ <https://pubs.acs.org/journal/jocean>

⁵ <https://cdnsiencepub.com/journal/cjc>

⁶ <https://onlinelibrary.wiley.com/journal/10991395>

⁷ <https://www.sciencedirect.com/journal/bioorganic-chemistry>

A Quick Collection of Papers

I spent 20 minutes looking at recent issues of ACS Journals⁸ and noticed the papers listed below. They may or may not be suitable, but they caught my eye.⁹ Now, I must skim them and make a shortlist. Then, I will read those papers and decide if there is data I can analyze and if there is a mystery I need to solve.

“Brønsted Acid-Catalyzed Reduction of Furans.” N. Frank, M. Leutzsch, B. List, *J. Am. Chem. Soc.*, **2025**, *147*, 7932-7938. <https://doi.org/10.1021/jacs.4c18485>

“Highly Acidic Electron-Rich Brønsted Acids Accelerate Asymmetric Pictet-Spengler Reactions by Virtue of Stabilizing Cation- π Interactions.” M.J. Scharf, N. Tsuji, et al., *J. Am. Chem. Soc.*, **2024**, *146*, 28339-28349. <https://doi.org/10.1021/jacs.4c09421>

“Concerted or Stepwise? An Experimental and Computational Study to Reveal the Mechanistic Change as a Result of the Substituent Effects.” X. Li, M.P. Meyer, *J. Org. Chem.*, **2025**, *90*, 2148-2155. <https://doi.org/10.1021/acs.joc.4c02197>

“DABCO-Catalyzed Synthesis of Thiazolidine-2-thiones: System Development and Mechanistic Insights.” S.G. Chalkidis, S. Hong et al., *J. Org. Chem.*, **2025**, *90*, 4302-4312. <https://doi.org/10.1021/acs.joc.5c00020>

“Bridging Machine Learning and Thermodynamics for Accurate pK_a Prediction.” W. Luo, G. Zhou et al., *JACS Au*, **2024**, *4*, 3451-3465. <https://doi.org/10.1021/jacsau.4c00271>

“Determination of the Hammett Acidity of HF/Base Reagents.” M. Longuet, K. Vitse et al., *J. Am. Chem. Soc.*, **2024**, *146*, 12167-12173. <https://doi.org/10.1021/jacs.4c02344>

“Neutral Phenolic Contaminants Are Not Necessarily More Resistant to Permanganate Oxidation Than Their Dissociated Counterparts: Importance of Proton-Coupled Electron Transfer.” T. Chen, H. Dong et al., *Env. Sci. Tech.*, **2023**, *57*, 17620-17628. <https://doi.org/10.1021/acs.est.3c05495>

“Remote Steric and Electronic Effects of N-Heterocyclic Carbene Ligands on Alkene Reactivity and Regioselectivity toward Hydrocupration Reactions: The Role of Expanded-Ring N-Heterocyclic Carbenes.” A.A. Hussein, A. Ariffin, *J. Org. Chem.*, **2023**, *88*, 13009-13021. <https://doi.org/10.1021/acs.joc.3c01121>

“Estimating the Bias of Model Compounds for the Determination of Species-Specific Protonation Constants.” T. Páll, Z. Kolompár et al., *ACS Omega*, **2024**, *9*, 896-902. <https://doi.org/10.1021/acsomega.3c06994>

“Linear Free Energy Relationships and Transition State Analysis of CO₂ Reduction Catalysts Bearing Second Coordination Spheres with Tunable Acidity.” K. Teindl, B.O. Patrick, E.M. Nichols, *J. Am. Chem. Soc.*, **2023**, *145*, 17176-17186. <https://doi.org/10.1021/jacs.3c03919>

⁸ Always use quality journals with a good reputation and (hopefully) an effective editor. Thousands of chemistry journals are available, and many of them are garbage. My bias is to consider any journal that starts with the word “international” unworthy of my time. If a nation won’t put its name in a journal, then that tells me something. Most countries are very proud of their names. Even good journals often publish bad papers. If the paper you are considering is poorly written, find another paper in that field and continue. Don’t try to write a good exploration based on a bad paper.

⁹ These papers are merely suggestions. Feel free to find a quality publication in an area of chemistry that interests you and critically analyze and interpret the subject. Investigate or apply the principles of physical organic chemistry as you do so.

A Quick Collection of Papers

Last year, I spent 20 minutes looking at recent issues of the Journal of Physical Organic Chemistry (and a few other journals) and noticed these papers. It took 20 minutes to find them, and now the work begins.

“A comparative study on the cucurbit[7]uril-based indicator displacement assay for methyl Viologen. A theoretical and experimental perspective.” P. Pavez, A. Fierro, et al., *J. Phys. Org. Chem.*, **2024**, 37, e4581. <https://doi.org/10.1002/poc.4581>

“Effect of Cucurbit[7]uril on contrasting binding with flavonoids: Insights from spectroscopy and calorimetric studies.” R. Jana, S. Pandit, et al., *J. Phys. Org. Chem.*, **2023**, 36, e4559. <https://doi.org/10.1002/poc.4559>

“The road to bis-periazulene (cyclohepta[*def*]fluorene): Realizing one of the long-standing dreams in nonalternant hydrocarbons.” A. Konishi, K. Horii, M. Yasuda, *J. Phys. Org. Chem.*, **2023**, 36, e4495. <https://doi.org/10.1002/poc.4495>

“Revisiting the formation mechanism of diarylamines via Smiles rearrangement.” F. Murillo, A. Quintal, et al., *J. Phys. Org. Chem.* **2022**, 35, e4427. <https://doi.org/10.1002/poc.4427>

“Solvolyses of naphthoyl chlorides. Solvent effect and Grunwald–Winstein correlation analyses with Y_{XBnCl} scales.” K.-T. Liu, P.Y. Hwang, H.I. Chen, *J. Phys. Org. Chem.*, **2002**, 15, 750-757. <https://doi.org/10.1002/poc.549>

“A design to prevent floating within the N scale of nucleophilicity.” T.W. Bentley, *J. Phys. Org. Chem.*, **2010**, 23, 836-844. <https://doi.org/10.1002/poc.1670>. see also <https://www.cup.lmu.de/oc/mayr/CDmayrPoster.html>

“Steric effects of alkyl groups: A ‘cone angle’ approach.” D. Datta, D. Majumdar, *J. Phys. Org. Chem.*, **1991**, 4, 611-617. <https://doi.org/10.1002/poc.610041005>

“Structure and reactivity in the hydrolyses of aliphatic carboxylic acid esters and chlorides.” A.C. Regan, C.I.F. Watt, *J. Phys. Org. Chem.*, **2007**, 20, 180-189. <https://doi.org/10.1002/poc.1144>

“Steric effect in alkylation reactions by N-alkyl-N-nitrosoureas: a kinetic approach.” J.A. Manso, M.T. Pérez-Prior, et al., *J. Phys. Org. Chem.*, **2008**, 21, 932-938. <https://doi.org/10.1002/poc.1402>

“Direct bromocarboxylation of arynes using allyl bromides and carbon dioxide.” Y. Zhang, W. Xiong, *Chem. Commun.*, **2019**, 55, 12304-12307, <http://dx.doi.org/10.1039/C9CC05495B>

“Understanding chemistry: from ‘heuristic (soft) explanations and reasoning by analogy’ to ‘quantum chemistry.’” J.I. Seeman, D.J. Tantillo, *Chem. Sci.*, **2022**, 13, 11461-11486, <https://doi.org/10.1039/D2SC02535C>

Do your own quick search and then pick through the pile for that diamond in the rough.